VICTORY 2027

Endocrine System

LECTURE-01 BIOLOGY

By : Upasana Ma'am



Topics to be Covered

1. Hormones / Glands
2. Pituitary Gland, Thyroid, Pancreas.
3. MAHA LIVE
4.


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上

RATTON KO KARO BYE BYE !!

EASY CONCEPTS SE KARO HELLO HI !!

With – Uppu Ma'am



Differences between Hormonal Control and Nervous Control

Endocrine System

Hormonal control	Nervous control
1. Usually slow.	1. Immediate/Rapid.
2. Transmitted chemically through blood.	2. Transmitted electrochemically through the nerve fibres and chemically across synapses.
3. Affects different organs (widespread in body).	3. Affects only the particular muscles or the gland (local).
4. Effect is short term or long-lasting.	4. Effect only short-lived.
5. Can affect growth.	5. Cannot affect growth.
6. Can bring about specific chemical changes and regulates metabolism.	6. Does not influence chemical changes and cannot regulate metabolism.
7. Cannot be modified by learning from previous experience.	7. Can be modified to some extent by learning from previous experience.

Glands → It can be a Cell, tissue, organ which secrete some chemicals or substances.

E.g → LIVER (largest gland), salivary gland

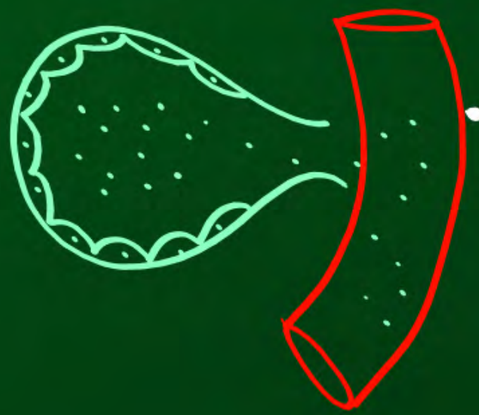
EXOCRINE GLAND

→ Duct glands

→ E.g Salivary gland, sweat gland, lacrimal gland, Sebaceous gland



ENDOCRINE GLAND



- Ductless gland.
- Secretions [HORMONES] are directly added in Blood.

ENDOCRINE GLANDS

- Releasing Hormone
- Inhibiting Hormone

Hypothalamus राजमाता





Pituitary

Thyroid

Pancreas

Adrenal gland (paired)

Anterior Pituitary

Posterior Pituitary

Bahubali

Somatotropin/

→ Growth Hormone
→ TSH
→ ACTH
→ GSH → FSH
→ LH

(Name - Iropin
, -stimulating)

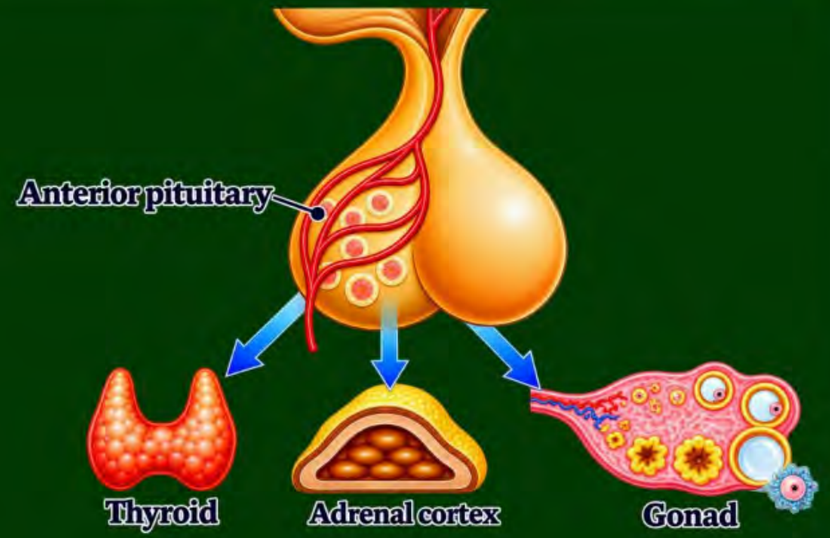
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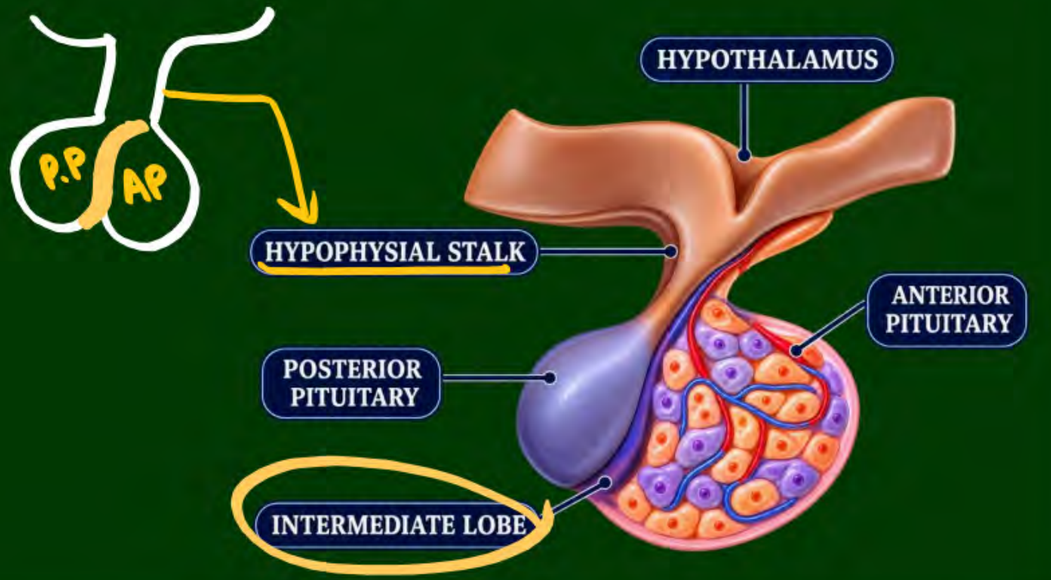
→ Oxytocin
→ ADH/Vasopressin

Pituitary

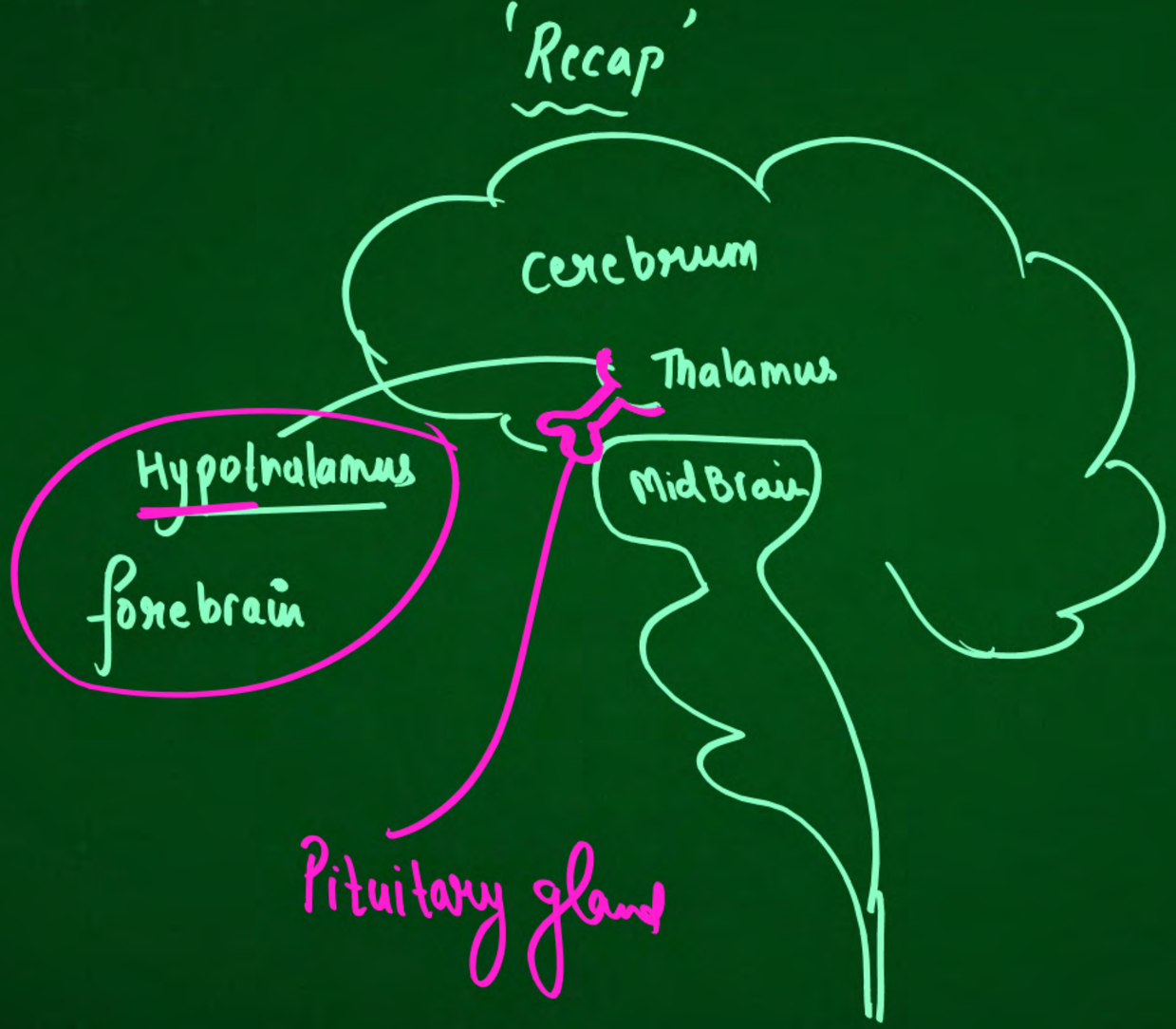
location → below Hypothalamus

- The pituitary gland is a small projection (about the size of a pea seed) which hangs from the base of the mid-brain below hypothalamus.
- It is called the master gland because it seems to control practically all other endocrine glands.
- Has two distinct lobes, the anterior pituitary and posterior pituitary (the entire pituitary gland weighs only about one-half gram).





'Recap'

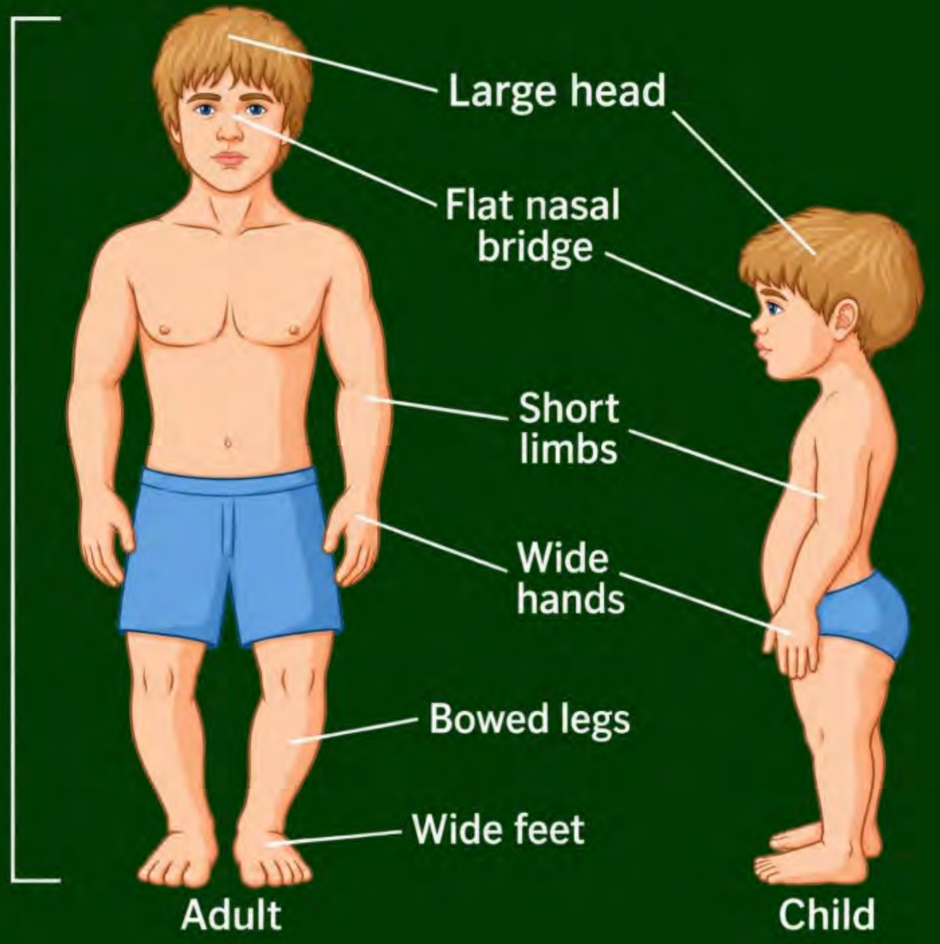

cerebrum
Thalamus
Hypothalamus
forebrain
MidBrain
Pituitary gland

Hormones from Anterior Pituitary

Growth Hormone:

Hyposecretion / under secretion

- Growth hormone (GH) is essential for normal growth. It is also called somatotropin (somatic: body, tropic: stimulating).
- The deficiency of GH in childhood results in dwarfism.



Hormones from Anterior Pituitary

- The over secretion in childhood results in gigantism; the long bones lengthen beyond normal and human giants up to 2.7 metres in height are produced.

Hypersecretion / over secretion



Hormones from Anterior Pituitary

- If over secretion of the growth hormone suddenly occurs in an adult, there is an excessive growth of bones in the face (particularly the jaws) and in the hands and feet.
- The person develops a large nose and thick lips.
- This condition is **acromegaly** (acro: height / extremity, megaly: large).



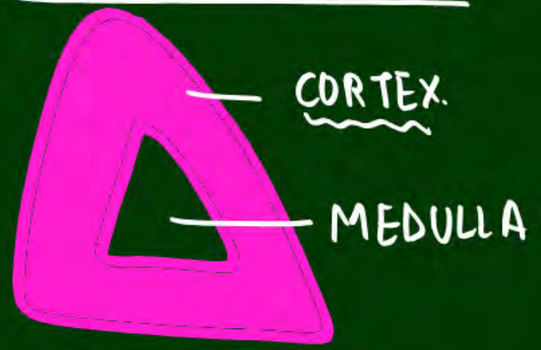
Hormones from Anterior Pituitary

- **THYROID STIMULATING HORMONE** (TSH) activates thyroid to secrete thyroxine
- **GONAD** (Reproductive Organs) **STIMULATING** (GONADOTROPINS) hormones regulate the activities of the testes and ovaries.

FSH — Follicle Stimulating Hormone

LH — Luteinizing Hormone

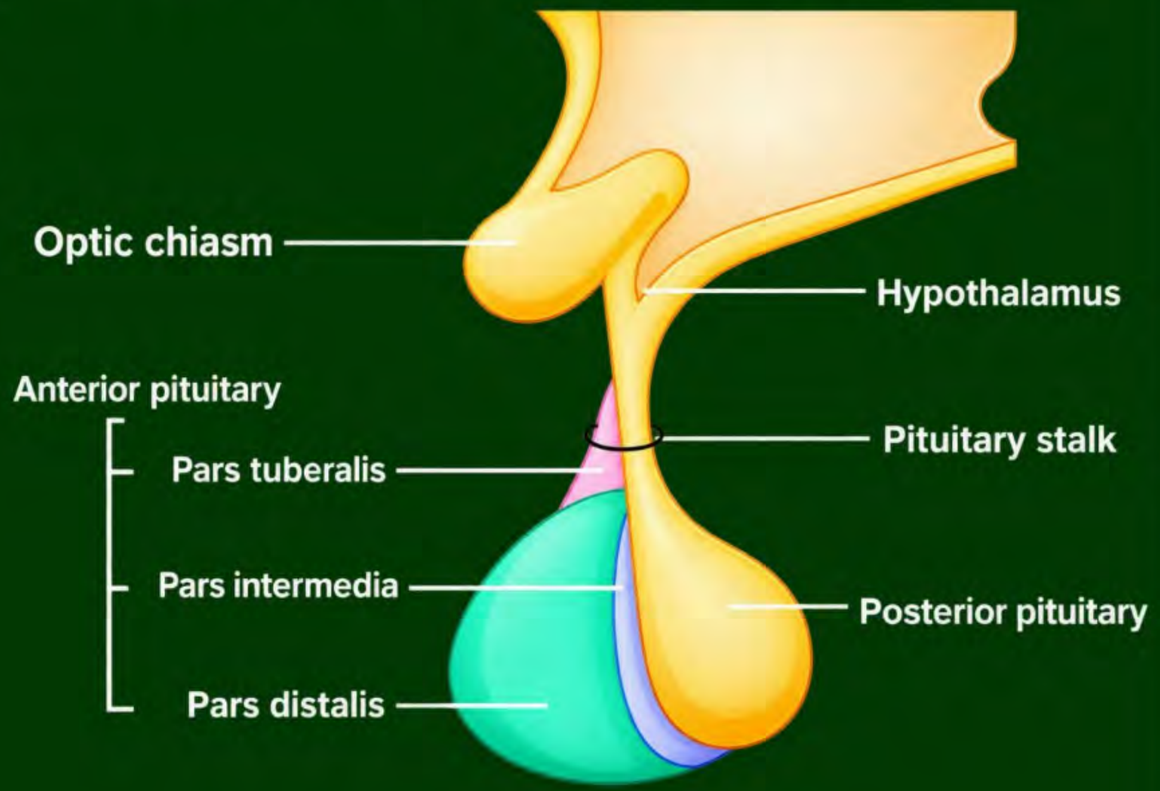
- **ADRENOCORTICOTROPIC HORMONE (ACTH)** regulates the activity of adrenal cortex.



Hormones from Posterior Pituitary

- The posterior lobe of the pituitary gland produces two hormones named vasopressin and oxytocin.

Pituitary gland anatomy

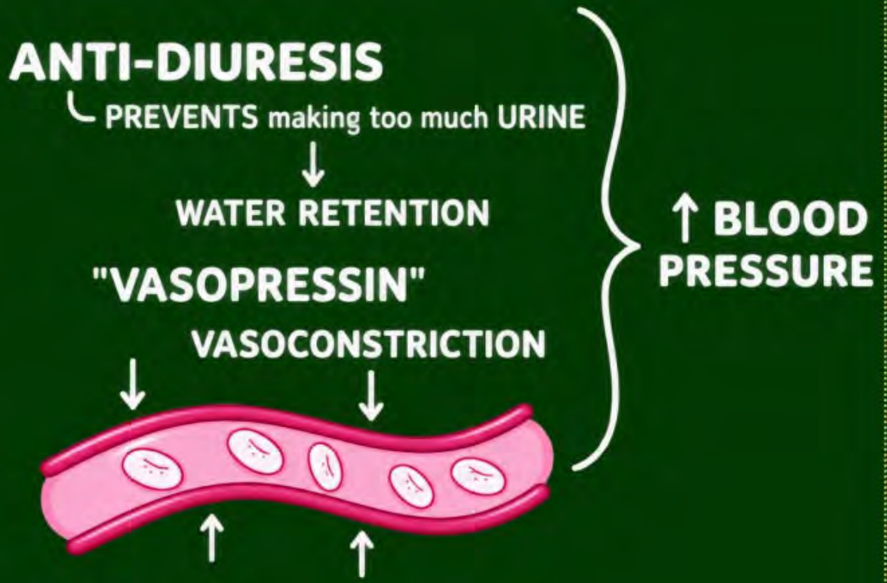


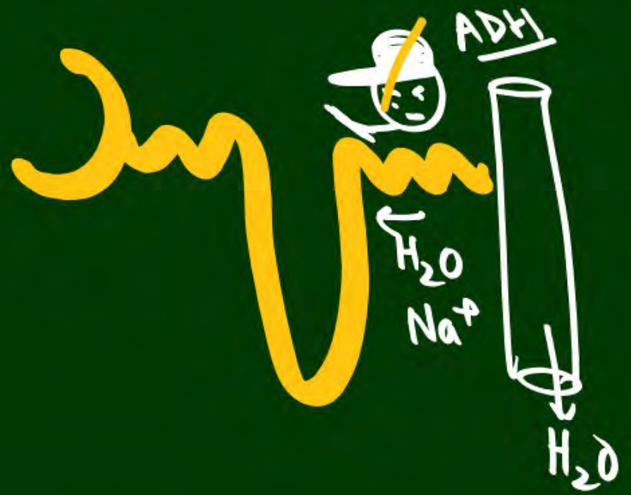
vaso → Blood vessel



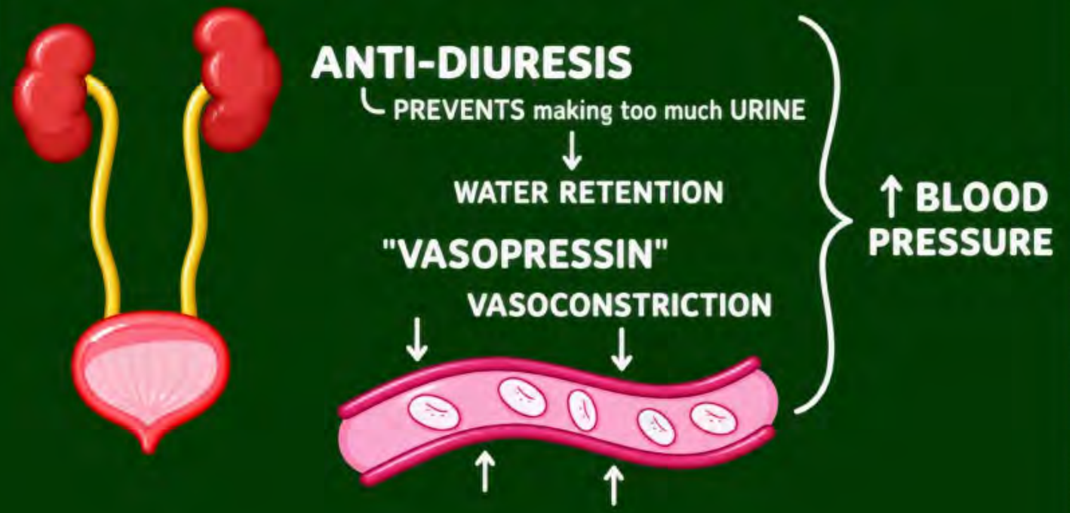
1. Antidiuretic hormone (ADH), also called vasopressin, constricts blood vessels with rise in blood pressure. It also acts on the kidney increasing reabsorption of water from the kidney tubules.



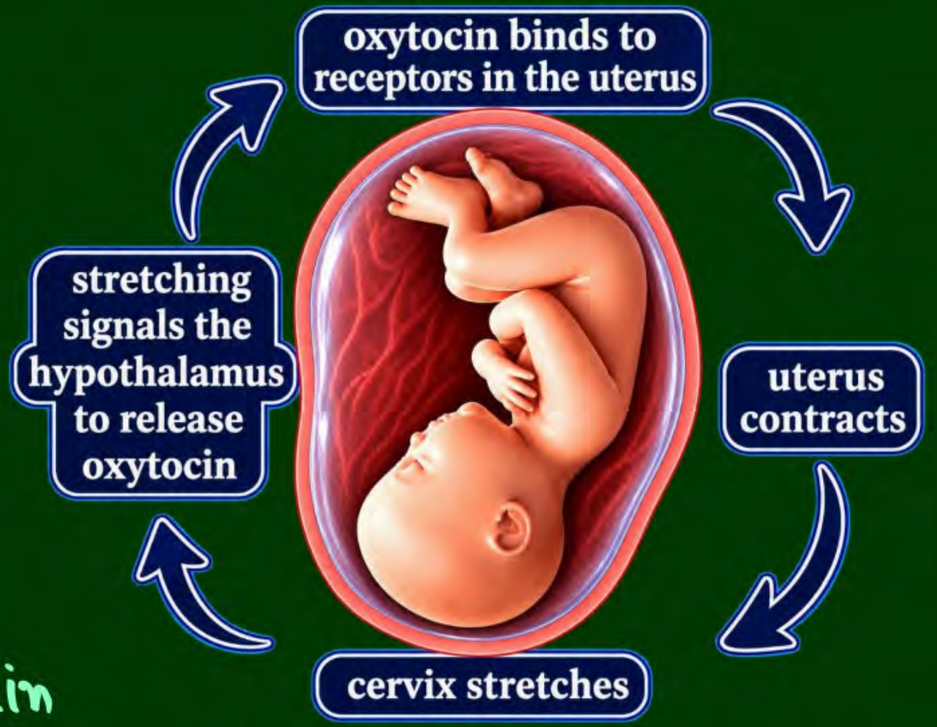




- Deficiency of ADH causes diabetes insipidus (water diabetes) in which urination is frequent and copious, resulting in loss of water from the body and the person becomes thirsty.



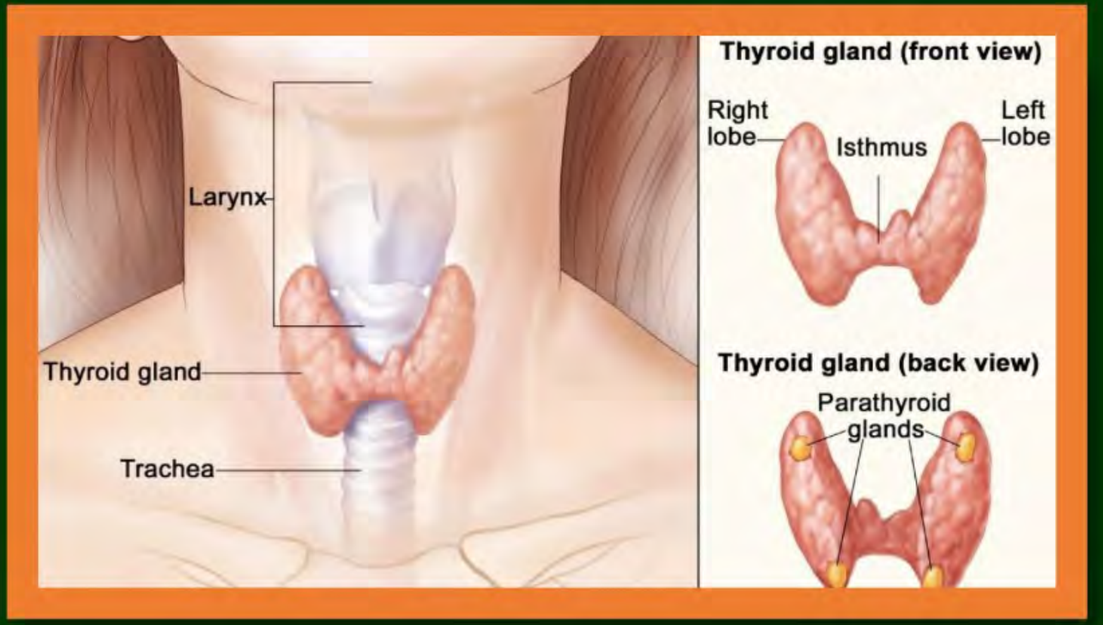
2. Oxytocin (Gk. oxys : sharp / quick, tokos : childbirth) stimulates vigorous contractions of the uterus in a pregnant mother, leading to the birth of the baby. It also stimulates milk ejection.

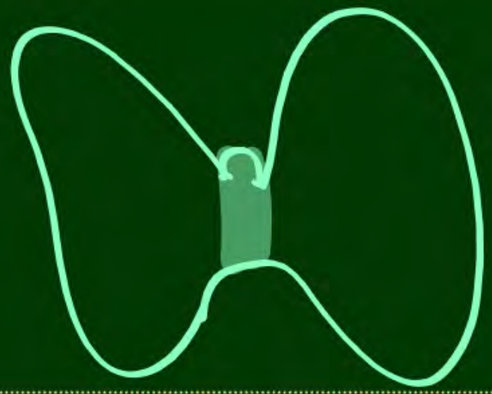


Milk Production → Prolactin

Thyroid Gland

- The thyroid is a bilobed (butterfly - shaped) structure situated in front of the neck just below the larynx.
- The two lobes are joined by a narrow isthmus (interconnection).
- It secretes two hormones thyroxine and calcitonin.





Which hormone stimulates Thyroid gland?

A) GnRH

B) TSH ✓

C) ACTH

D) GH

Which hormone stimulates Adrenal Cortex. ?

A) GnRH

B) TSH

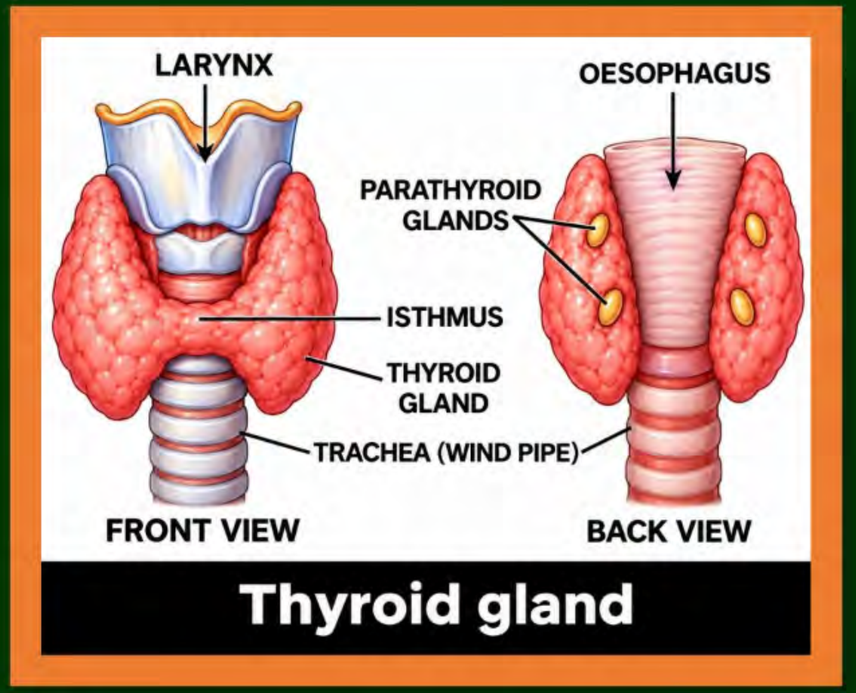
C) ACTH ✓

D) GH



B.P↑

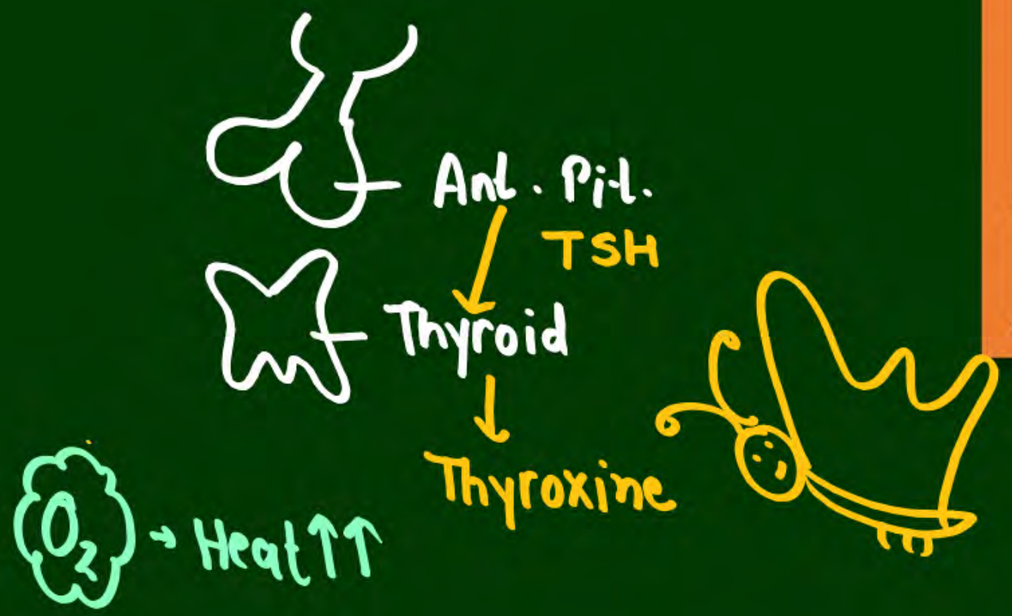
- Thyroxine regulates the basal metabolism, i.e., the rate of cellular oxidation resulting in heat production at rest.
- An increase in the secretion increases metabolism and a decrease in secretion lowers it.

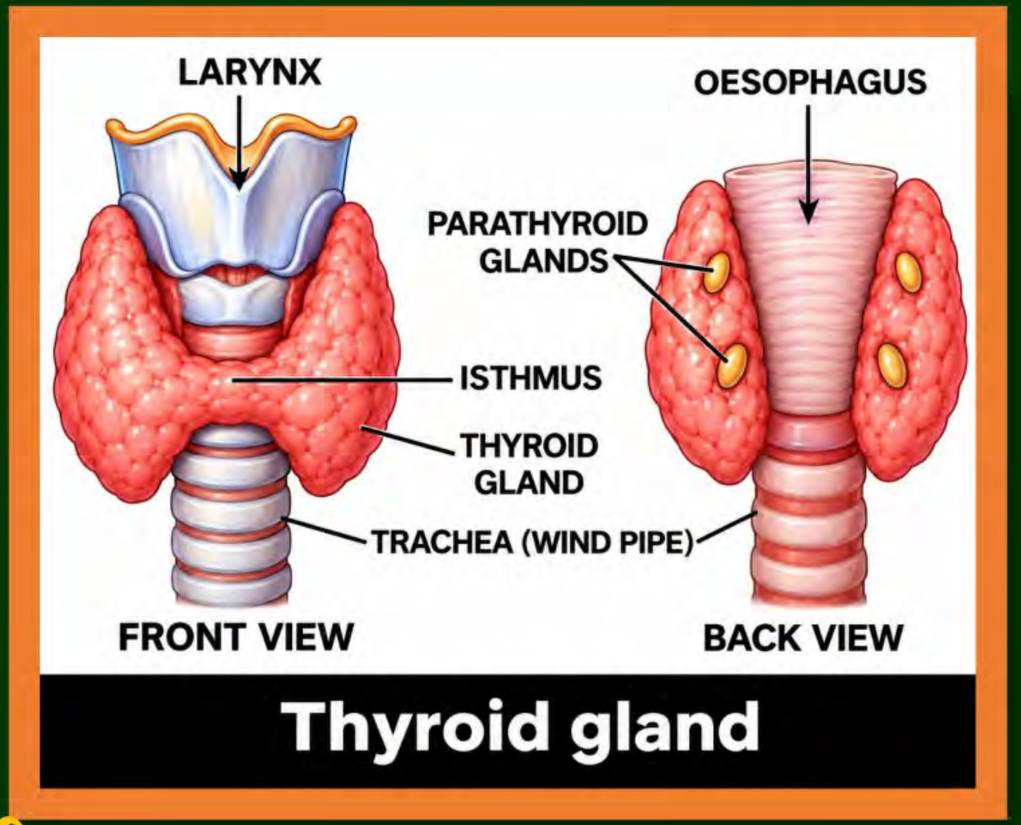


BMR → Basal Metabolic Rate

At Rest Glucose + O_2 → ATP | Energy ✓ · Heat ↑

- It also influences the general growth of the body, ossification of bones, body temperature, mental development, etc.

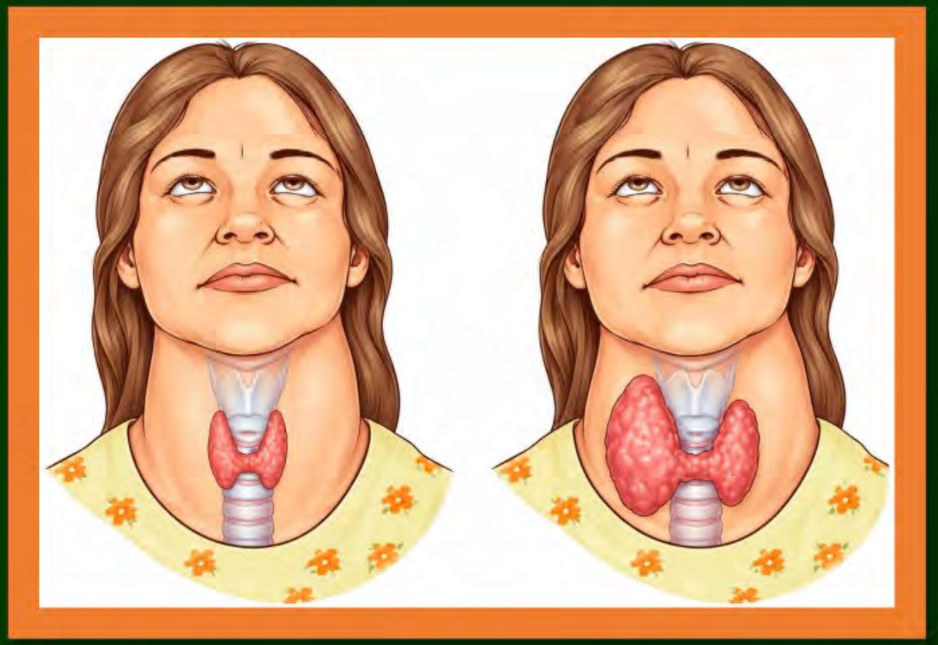




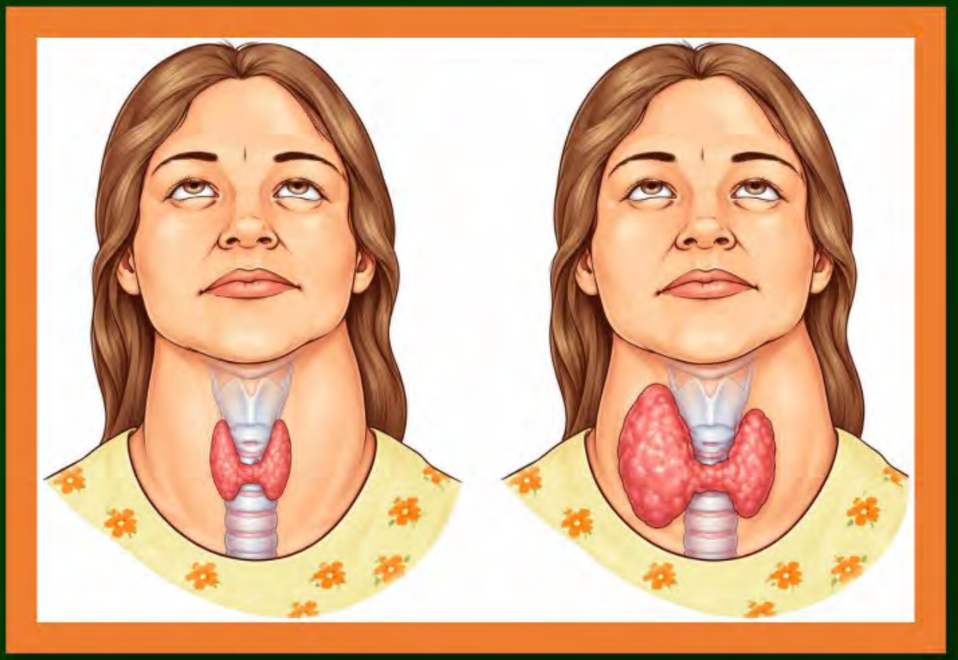
A. **Undersecretion (Hypothyroidism) :**

Insufficient secretion of thyroxine may lead to three conditions - simple goitre, cretinism and myxoedema.

(i) Simple goitre is the enlargement of the thyroid and is visible as a swelling in the neck.



- This is due to insufficient quantity of iodine in food.
- This condition is common in the people living in hilly regions where iodine is deficient in soil and hence in the food grown there.



(ii) Cretinism is a condition which affects the growth of children showing dwarfism and mental retardation.

This is due to defective development, or early atrophy (degeneration) of the thyroid.





swelling Gull's disease.

(iii) Myxoedema is a condition that affects an adult if his thyroid does not function properly;
in this condition, the person becomes sluggish with swelling of the face and hands.



Q. Which hormone hypersecretion causes Acromegaly?

A) ACTH

B) Thyroxine

C) Growth Hormone ↑↑

D) Oxytocin.

B. Oversecretion (Hyperthyroidism) :

Excess of thyroxine secretion may also cause a kind of goitre called **exophthalmic goitre** (exo : outward, ophthalmos: eye).



- A person having the problem of over - secretion shows:



- Protrusion of eyes,
- Formation of goitre in the neck,
- A marked increase in metabolic rate,
- Rapid heart-beat, and
- Shortness of breath.

QUESTION

Fill in the blanks by selecting the appropriate alternatives from those given in brackets.

Regulation of activities inside the body is brought about by the _____ and _____ systems.

Options:

A Nervous

B Digestive

C Endocrine

QUESTION

(ii) Hormones are secreted directly into the _____.

A Organs

B Lymph

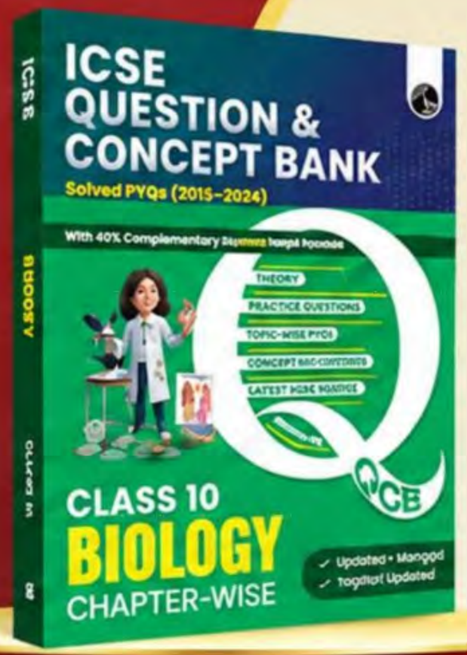
C Blood

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